Open-Ended Laboratory Problem (Washington Accord – Complex Engineering Problem)

Title: *Photolithography-Based Fabrication and Process Optimization of an n-MOS Device under Performance, Safety, and Environmental Constraints*

Problem Statement (CEP Formulation)

The fabrication of **n-MOS transistors** for low-power electronic systems requires the integration of **semiconductor physics, materials science, and microfabrication processes**. In a laboratory environment, achieving acceptable device quality using **photolithography** is complicated by **process variability, material limitations, safety risks, and environmental impacts of chemicals and waste**.

You are required to **address a complex engineering problem** by designing, fabricating, and evaluating an **n-MOS device using photolithography**, where **multiple interdependent process parameters** must be selected and optimized under **conflicting technical, safety, and environmental constraints**. The problem does not have a single correct solution and requires **engineering judgment, analysis, and justification**.

Engineering Tasks (CEP-Oriented)

1. **Problem Scoping and Requirement Identification**
    - Identify functional requirements of a laboratory-scale n-MOS device suitable for low-power operation.
    - Identify technical, safety, and environmental constraints influencing the fabrication process.
2. **Process Design Using Modern Engineering Tools (PO-e)**
    - Design a complete **n-MOS fabrication flow** using photolithography, specifying:
        - Photoresist parameters
        - Exposure and development conditions
        - Gate oxide thickness and gate geometry
    - Justify the selection of fabrication parameters using engineering principles.
3. **Implementation and Observation**
    - Execute photolithography and thin-film processing steps using modern laboratory tools.
    - Observe and document pattern quality, alignment accuracy, and fabrication limitations.
4. **Analysis of Interdependent Parameters**
    - Analyze how variations in photolithography resolution, oxide thickness, and alignment accuracy collectively influence:
        - Threshold voltage
        - Leakage current
        - Device reliability
    - Demonstrate understanding of interactions among multiple process variables.
5. **Safety and Environmental Risk Evaluation**

    - Identify hazards related to UV exposure, photoresist chemicals, solvents, and waste.
    - Evaluate risk mitigation strategies and environmentally responsible waste-handling practices.
6. **Trade-Off Analysis and Optimization**
    - Propose process modifications to improve device performance or reliability while reducing safety risks or environmental impact.
    - Justify trade-offs where improvement in one aspect leads to constraints in another.

Washington Accord – CEP Attributes Addressed

CEP Attribute	Demonstration in the Problem
In-depth knowledge	Semiconductor devices, photolithography, materials
Range of issues	Technical performance, safety, environmental impact
Conflicting constraints	Device quality vs. chemical usage vs. safety
Use of modern tools	Photolithography systems, thin-film deposition, microscopy
Non-routine solution	Multiple valid fabrication and optimization paths
Engineering judgment	Parameter selection and trade-off justification

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Expected Learning Outcome (CEP & PO-e)

Students will demonstrate the ability to **formulate and solve a Complex Engineering Problem** by:

- Integrating **theory and modern fabrication tools**
- Managing **interacting technical and non-technical constraints**
- Making **informed engineering judgments** in the absence of a single correct solution

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PO(e) Attainment Statement

This laboratory activity enables students to select and apply appropriate modern microfabrication and characterization tools to formulate, analyze, and address a complex engineering problem involving n-MOS device fabrication under realistic constraints.

<u>Assessment Rubrics (Total = 10 Marks)</u>

Assessment Rubrics with CO–PO Mapping

Course Outcome (CO):

- **CO2:** Fabricate and analyze basic MOS devices using photolithography and thin-film processes

Program Outcome (PO):

- **PO(e):** Modern Tool Usage

Rubric Table (10 Marks)

Assessment Criteria	CO	PO	Excellent	Average	Marginal / Poor	Marks
1. Problem Formulation & Process Planning	CO2	PO(e)	Clearly formulates the fabrication problem as a complex engineering problem; identifies constraints and plans complete n-MOS process flow	Problem identified with partial planning and limited consideration of constraints	Problem poorly formulated; incomplete or incorrect process flow	**2**
2. Use of Photolithography & Modern Tools	CO2	PO(e)	Correct and effective use of spin coater, mask aligner, UV exposure, developer with justified parameters	Tools used correctly but without proper parameter justification	Incorrect or unsafe tool usage	**2**
3. Pattern Quality & Fabrication Accuracy	CO2	PO(e)	Well-defined gate pattern with good alignment and minimal defects	Pattern formed with minor defects or misalignment	Poor or incomplete pattern formation	**2**
4. Analysis of Process–Performance Relationship	CO2	PO(e)	Clearly analyzes how lithography resolution and oxide thickness affect n-MOS behavior	Basic qualitative analysis with limited linkage	Little or no meaningful analysis	**2**
5. Safety & Environmental Considerations	CO2	PO(e)	Identifies major hazards and applies proper safety and waste-management practices	Identifies hazards but safety/environmental practices are incomplete	Poor awareness of safety and environmental issues	**1**
6. Engineering Judgment & Improvement Proposal	CO2	PO(e)	Proposes realistic process improvements with justified trade-offs	Suggests improvements with weak justification	No improvement or unjustified suggestions	**1**

CO–PO Attainment Statement

This assessment evaluates students' ability to fabricate and analyze an n-MOS device by selecting and applying modern photolithography and thin-film tools, addressing a complex engineering problem involving interacting technical, safety, and environmental constraints, thereby achieving CO2 and PO(e).